

Learning with Less: Label-Efficient Land Cover Classification at Very High Resolution

Author Order?

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- VHSR (<5 m) land use/land cover products reveal fine-scale features critical for many applications ().
- Modern CNN/ViT models are more than capable of crossing the “resolution barrier” that plagued earlier methods ().

Practical barriers land cover mapping with semantic segmentation at VHSR

- DL models are powerful but data-hungry (Zhu et al., 2017).
- Most research into VHSR land cover mapping use benchmark datasets with abundant labels over small areas (ISPRS, 2018b,a)
- Creating large, labeled datasets for new areas is a major bottleneck in terms of human capital

Two-part approach to operational VHSR land cover mapping with few labels

How can we train accurate models with limited labeled data?

- **Part 1:** Sample smarter. Use existing, coarse land cover data to guide selection of a small but informative training set based on approximate composition of candidate patches.
- **Part 2:** Pre-train better. Use self-supervised learning on large amounts VHSR imagery to learn robust, domain-specific features without labels.

Part 1: Smarter Sampling with Semantic Stratification

- **Idea:** Don't just sample randomly. Select patches that are representative of the diverse land cover compositions in the study area.
- **Method:** Use existing, coarse land cover data (like NLCD) to guide our sampling.

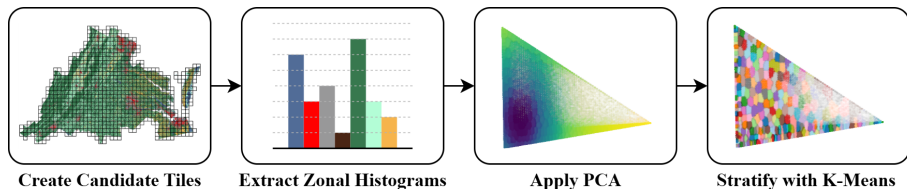


Figure: We cluster candidate patches based on their land cover mix (from NLCD) and sample from these clusters.

Semantic Stratification: The Result

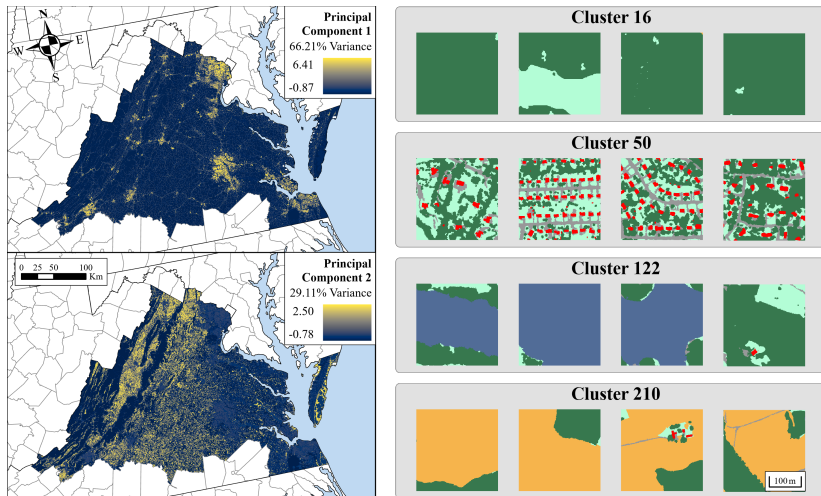


Figure: Patches within the same cluster tend share land cover characteristics.

Validation: Does it Work?

Yes! Semantic stratification leads to better models.

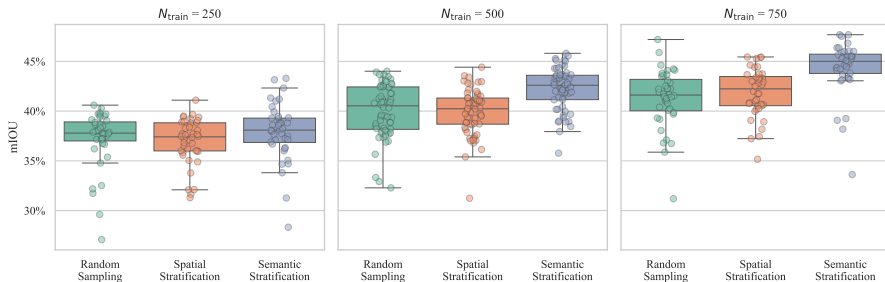


Figure: Our proposed method shows statistically significant performance gains over random sampling and spatially stratified sampling at $N_{\text{train}} = 500$ and $N_{\text{train}} = 750$.

Part 2: Better Pre-training with Self-Supervision

- **Idea:** Leverage vast amounts of *unlabeled* imagery to learn meaningful features before fine-tuning on our small labeled dataset.
- **Method:** Use Bootstrap Your Own Latent (BYOL), a self-supervised learning technique (Grill et al., 2020).

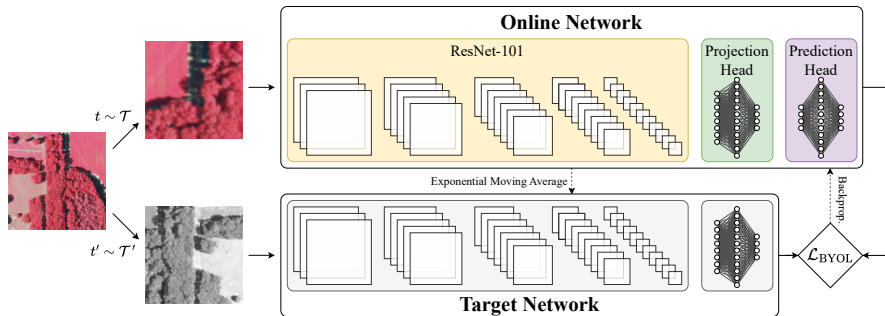


Figure: BYOL trains a network to predict a different augmented view of the same image, learning robust, transformation-invariant features.

Validation: Are the Features Better?

Yes! The learned features are far more effective for our task.

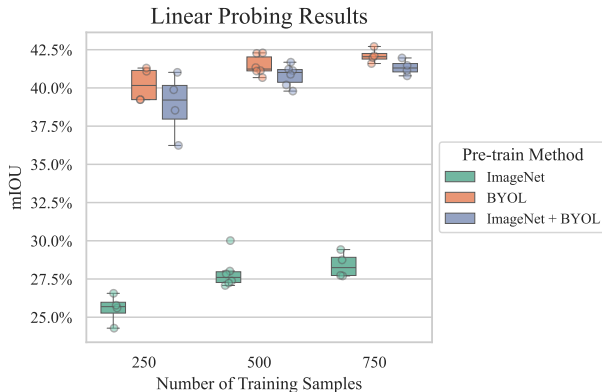


Figure: Linear probing shows features from BYOL models vastly outperform standard ImageNet features, closing the domain gap.

Impact on Fine-Tuning Performance

This translates to better performance on the final segmentation task.

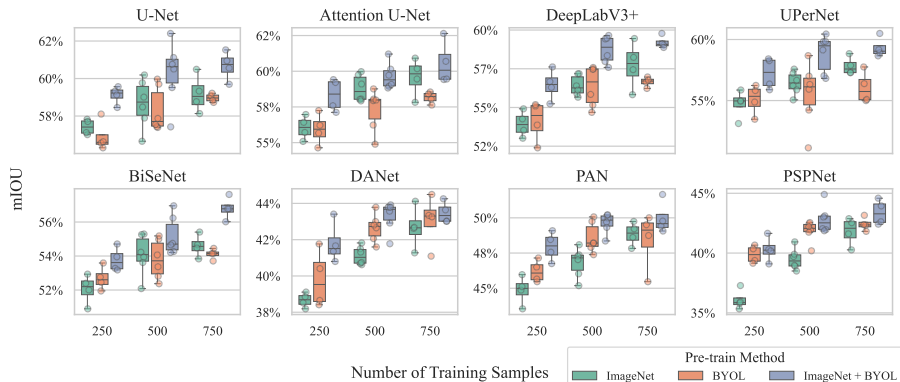
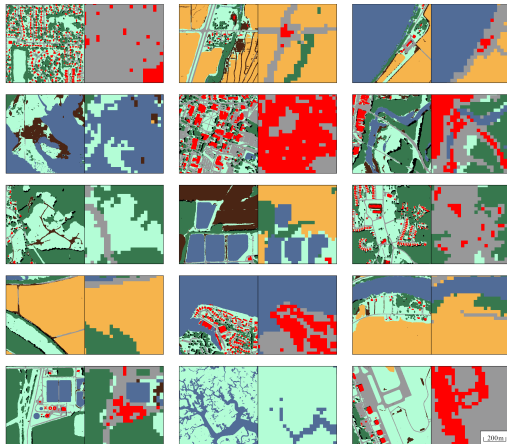
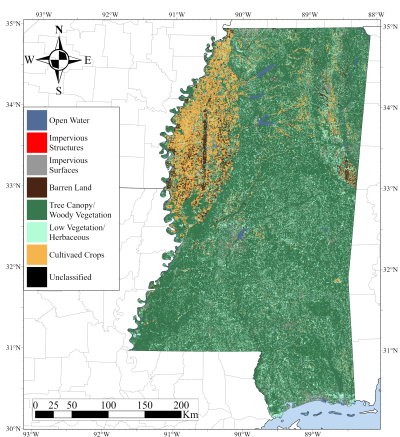


Figure: Fine-tuning various segmentation models shows that BYOL pre-training consistently improves mIoU, especially with few labeled samples.

Final Result: A 1m Land Cover Map of Mississippi

- Using our two-part strategy, we trained an ensemble of models with only 1,000 labeled samples.
- This produced the first publicly-available, 1-meter resolution land cover map for the entire state.



Visual Assessment

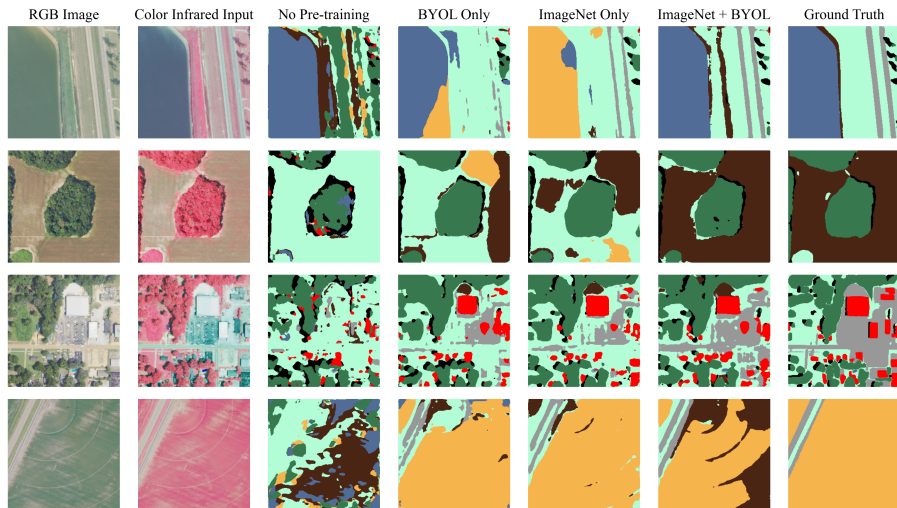


Figure: The model accurately captures fine-scale features like individual buildings, roads, and vegetation boundaries.

- High-quality, large-area VHSR land cover mapping is possible with very limited labeled data.
- **Key Takeaways:**
 - ① **Sample Smarter:** Semantic stratification using existing coarse data is a powerful way to build more informative training sets.
 - ② **Pre-train Better:** Self-supervised learning on unlabeled domain-specific imagery is highly effective at bridging the domain gap.
- This work provides a blueprint for label-efficient deep learning in remote sensing.

Questions?

References I

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